

WSGI

WSGI is the web server gateway interface between web servers and web apps for Python. It is a specification that describes how a web server communicates with web applications, and how web applications can be chained together to process one request.

WSGI forwards requests from users received by the web server to the Python script. The output of the Python script is forwarded to WSGI, then to the web server, and finally to the user

acting as an interface between the front-end server (Apache) and the Flask application (refer Figure 1).

mod_wsgi

This Apache HTTP server module provides a WSGI-compliant interface for hosting Python based web applications under Apache. The mod_wsgi module embeds the Python application (the Flask app) within the server, and allows communication via the Python WSGI interface.

 **Note:** We will be using Python3.6 for development, so we should install *mod_wsgi* (py3).

Setting up the environment

We will start by installing the Apache 2 server. Use the following command:

```
$sudo apt install apache2
```

We will use *systemctl*, a utility for controlling the *systemd* system and service manager; it is used to start, restart, stop services and beyond. The *systemctl status* sub-command, as the name states, is used to view the status of a service. You can use it for the above purpose by using the following command:

```
$sudo systemctl status apache2
```

The output of the command will confirm the status of Apache and its installation.

After letting the command run, all required packages are installed, and we can test this out by typing in our IP address in the web browser. If the Apache server is running properly, you will see the 'Apache2 Ubuntu Default Page'.

Installing and setting up Flask

To install Flask, you should be in your applications root directory. Flask, its dependencies, and most Flask extensions support Python 3.

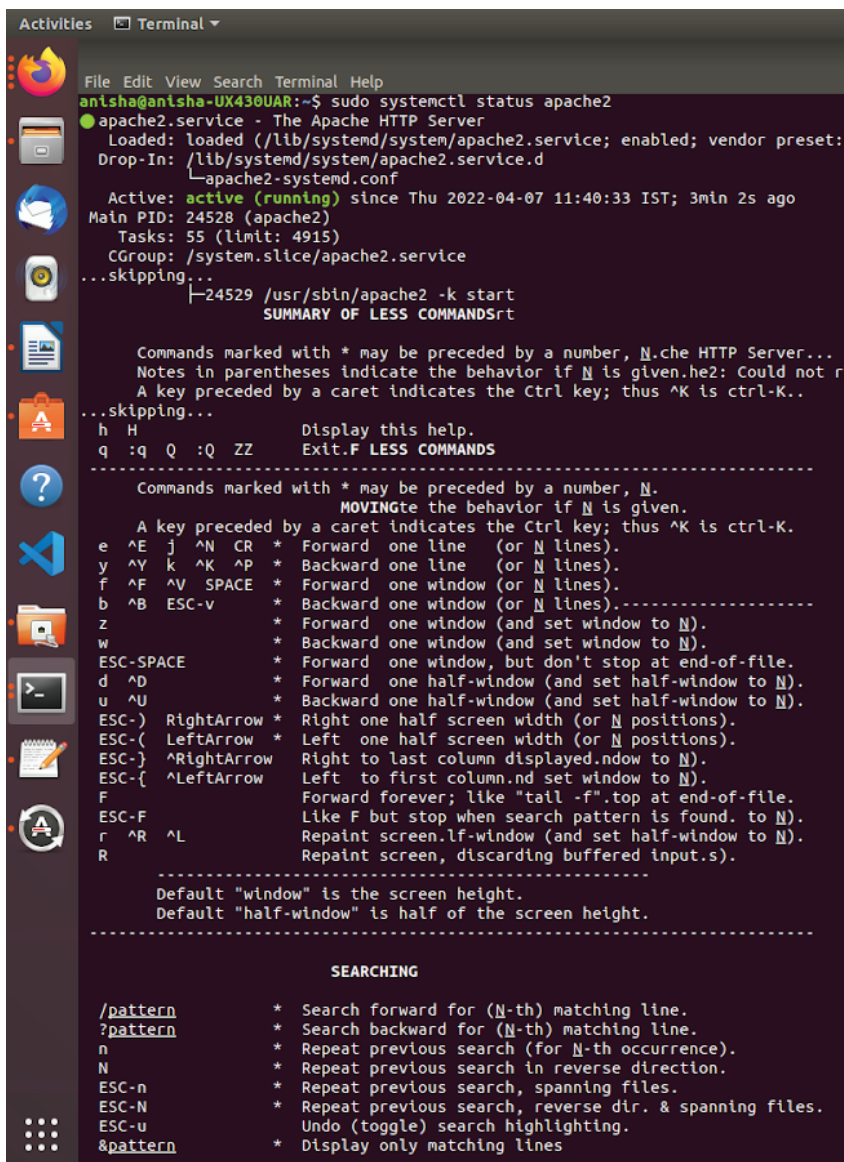
The standard package manager for Python is *pip*. It allows you to install and manage packages that aren't part of the Python standard library.

Now we will install Python3 in the next step using the following command:

```
$sudo apt install python3
```

```
$sudo apt install python3-pip
```

Python3 mainly affects WSGI middleware interacting with the WSGI environ data. Werkzeug, a comprehensive WSGI web application



```

anisha@anisha-UX430UAR:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset:
  Drop-In: /lib/systemd/system/apache2.service.d
           └─apache2-systemd.conf
   Active: active (running) since Thu 2022-04-07 11:40:33 IST; 3min 2s ago
     Main PID: 24528 (apache2)
       Tasks: 55 (limit: 4915)
    CGroup: /system.slice/apache2.service
            ...skipping...
            └─24529 /usr/sbin/apache2 -k start
                SUMMARY OF LESS COMMANDSrt

Commands marked with * may be preceded by a number, N, che HTTP Server...
Notes in parentheses indicate the behavior if N is given.he2: Could not r
A key preceded by a caret indicates the Ctrl key; thus ^K is ctrl-K..
...skipping...
h H          Display this help.
q :q Q :Q ZZ Exit.F LESS COMMANDS
-----
Commands marked with * may be preceded by a number, N.
MOVINGte the behavior if N is given.
A key preceded by a caret indicates the Ctrl key; thus ^K is ctrl-K.
e ^E j ^N CR * Forward one line (or N lines).
y ^Y k ^K ^P * Backward one line (or N lines).
f ^F ^V SPACE * Forward one window (or N lines).
b ^B ESC-v    * Backward one window (or N lines).-----
z            * Forward one window (and set window to N).
w            * Backward one window (and set window to N).
ESC-SPACE   * Forward one window, but don't stop at end-of-file.
d ^D        * Forward one half-window (and set half-window to N).
u ^U        * Backward one half-window (and set half-window to N).
ESC-) RightArrow * Right one half screen width (or N positions).
ESC-( LeftArrow  * Left one half screen width (or N positions).
ESC-) ^RightArrow * Right to last column displayed.ndow to N).
ESC-[ ^LeftArrow  * Left to first column.nd set window to N).
F            * Forward forever; like "tail -f".top at end-of-file.
ESC-F        * Like F but stop when search pattern is found. to N).
r ^R ^L       * Repaint screen.lf-window (and set half-window to N).
R            * Repaint screen, discarding buffered input.s).
-----
Default "window" is the screen height.
Default "half-window" is half of the screen height.
-----
SEARCHING
/pattern    * Search forward for (N-th) matching line.
?pattern    * Search backward for (N-th) matching line.
n           * Repeat previous search (for N-th occurrence).
N           * Repeat previous search in reverse direction.
ESC-n       * Repeat previous search, spanning files.
ESC-N       * Repeat previous search, reverse dir. & spanning files.
ESC-u       * Undo (toggle) search highlighting.
&pattern    * Display only matching lines
  
```

Figure 2: Apache status